

Policy & Leave Concierge

An enterprise HR assistant that answers employee questions about policies and leave balances with zero hallucination, grounded in live database records and official PDF documents.

The Problem

HR teams at HC Services face a common challenge: employees repeatedly ask the same policy questions - How many leave days do I have? Who approves expenses over 5,000 AED? Can I work remotely during probation? - and getting accurate, personalized answers requires cross-referencing:

- The employee's live HR records (leave balances, manager, probation status)
- The relevant policy documents (leave rules, approval thresholds, eligibility criteria)

Standard chatbots hallucinate. They might say "you have 20 days remaining" when the employee actually has 12, or cite policy clauses that don't exist.

HCS-01 solves this by enforcing dual-source grounding: every answer combines exact SQL database facts with semantically retrieved policy context, and every response shows precisely which sources were used.

How It Works

- **Employee asks a question**
In English or Arabic - the system auto-detects language and responds in kind.
- **SQL Database Lookup**
Fetch the employee's exact profile, leave balances, manager history, and recent requests from the relational database. This is ground truth.
- **Vector Search on Policies**
Embed the query and retrieve the most relevant chunks from official HR policy PDFs - including transcribed flowcharts and decision trees.
- **Prompt Assembly**
Inject the SQL facts and policy context into a structured system prompt. The LLM sees both the employee's actual data and the relevant policy rules.
- **Response with Citations**
The answer arrives with two types of sources: a database citation and PDF policy citations with page-level deep-links.

Key Features

DUAL-SOURCE GROUNDING

Combines deterministic SQL facts (leave balances, manager name) with semantic policy retrieval (RAG over PDF

documents).

MULTIMODAL DOCUMENT UNDERSTANDING

Policy PDFs contain flowcharts and decision trees. The system extracts and transcribes these visual elements.

BILINGUAL (ENGLISH & ARABIC)

Multilingual embeddings enable cross-lingual retrieval: an Arabic question retrieves English policy chunks.

CONVERSATION MEMORY

LangChain-powered session history enables multi-turn conversations with context carryover.

Architecture

Backend

- FastAPI
- LiteLLM (Gemini Flash)
- LangChain memory
- Qdrant vector DB
- SQLAlchemy + SQLite
- PyMuPDF

Frontend

Data Model

TABLE	PURPOSE
employees	Profile, role, department, manager, probation status
leave_balances	Entitled, used, remaining days by leave type
manager_history	Historical record of manager changes
leave_requests	Past requests with status and approver
expense_claims	Submitted claims with amounts and approval status

Policy Documents

REFERENCE	POLICY	TOPICS
HC-PC-001	Annual Leave	Entitlement, accrual, carry-over
HC-PC-002	Sick Leave	Allowance, certificates, Bradford Factor
HC-PC-003	Probation	Duration, reviews, benefits
HC-PC-004	Remote Work	Hybrid schedule, eligibility, core hours
HC-PC-005	Expense Claims	Approval tiers, per diem, receipts

Example Interaction

EMPLOYEE QUESTION

How many annual leave days do I have left, and who needs to approve if I take 5 days off?

CONCIERGE RESPONSE

You have 18 days of annual leave remaining this year. For a 5-day absence, you need to submit your request at least 10 business days in advance. Your line manager Ahmed Al Farsi will review team coverage, then HR validates against your balance.

SOURCES CITED

SQL Database: Employee EMP001 - 18 days remaining
HC-PC-001: Annual Leave Policy, Section 2.1 (Notice Periods) - Page 2